

Thm. Cauchy Integral Formula.

Suppose f is Holomorphic on some open set containing a closure of disk D .
If C is a Curve with Positive orientation of ∂D , then for any $z_0 \in D$,

$$f(z_0) = \frac{1}{2\pi i} \cdot \int_C \frac{f(z)}{z - z_0} dz$$

Proof. Let $z_0 \in D$ be fixed, and consider a Curve Γ_{ϵ} as
Since $\frac{f(z)}{z - z_0}$ is Holomorphic on $D \setminus \{z_0\}$, therefore



$$\int_{\Gamma_{\epsilon}} \frac{f(z)}{z - z_0} dz = 0. \text{ Taking } \epsilon \rightarrow 0, \text{ then we obtain}$$

$$\lim_{\epsilon \rightarrow 0} \int_{\Gamma_{\epsilon}} \frac{f(z)}{z - z_0} dz = \int_{C_{\epsilon}} \frac{f(z)}{z - z_0} dz + \int_C \frac{f(z)}{z - z_0} dz = 0.$$

Meanwhile, $\frac{f(z)}{z - z_0} = \frac{f(z) - f(z_0)}{z - z_0} + \frac{f(z_0)}{z - z_0}$ and since f is Holomorphic,
the first term is bounded, therefore $\lim_{\epsilon \rightarrow 0} \int_{C_{\epsilon}} \frac{f(z) - f(z_0)}{z - z_0} dz = 0.$

$$\begin{aligned} \text{Now, } \lim_{\epsilon \rightarrow 0} \int_{C_{\epsilon}} \frac{f(z_0)}{z - z_0} dz &= f(z_0) \cdot \lim_{\epsilon \rightarrow 0} \int_{C_{\epsilon}} \frac{1}{z - z_0} dz \\ &= f(z_0) \cdot \lim_{\epsilon \rightarrow 0} \left(\int_0^{2\pi} \frac{1}{\epsilon \cdot e^{it}} \cdot \epsilon \cdot i \cdot e^{it} dt \right) \\ &= -f(z_0) \cdot 2\pi i. \end{aligned}$$

$$\text{Consequently, } \int_C \frac{f(z)}{z - z_0} dz = 2\pi i f(z_0).$$

Corollary. $\int_C \frac{f(z)}{(z - z_0)^{n+1}} dz = 2\pi i \cdot \frac{f^{(n)}(z_0)}{n!}.$

Proof. Using induction for n . For $n=0$, clear. If $f^{(n-1)}(z_0) = \frac{(n-1)!}{2\pi i} \cdot \int_C \frac{f(z)}{(z - z_0)^n} dz$, then

$$\begin{aligned} \frac{f^{(n-1)}(z_0 + h) - f^{(n-1)}(z_0)}{h} &= \frac{(n-1)!}{2\pi i} \cdot \int_C f(z) \cdot \frac{1}{h} \left[\frac{1}{(z - z_0 - h)^n} - \frac{1}{(z - z_0)^n} \right] dz \\ &= \frac{(n-1)!}{2\pi i} \cdot \int_C f(z) \cdot \frac{1}{h} \left[\frac{h}{(z - z_0 - h)(z - z_0)^n} \cdot \left(\sum_{k=0}^{n-1} \frac{1}{(z - z_0 - h)^{n-1-k}} \cdot \frac{1}{(z - z_0)^k} \right) \right] dz \\ &\rightarrow \frac{(n-1)!}{2\pi i} \cdot \int_C f(z) \cdot \frac{1}{(z - z_0)^2} \cdot \sum_{k=0}^{n-1} \frac{1}{(z - z_0)^{n-1-k}} dz \quad \text{as } h \rightarrow 0 \\ &= \frac{(n-1)!}{2\pi i} \cdot \int_C f(z) \cdot n \cdot \frac{1}{(z - z_0)^{n+1}} dz \\ &= \frac{n!}{2\pi i} \cdot \int_C \frac{f(z)}{(z - z_0)^{n+1}} dz = f^{(n)}(z_0). \end{aligned}$$

Thm. Cauchy inequality.

$$|f^{(n)}(z_0)| \leq \frac{n!}{R} \cdot \sup_C |f| \text{ if } f \text{ is Holomorphic in an open containing } \overline{B_R(z_0)}, C = \partial B_R(z_0).$$

Thm. Suppose f is Holomorphic on an open Ω containing a closed disk $\overline{D}$ having center z_0 ,

$$f(z) = \sum_{n=0}^{\infty} \frac{f^{(n)}(z_0)}{n!} \cdot (z - z_0)^n \quad (z \in D)$$

Proof.
$$f(z) = \frac{1}{2\pi i} \cdot \int_C \frac{f(\mu)}{\mu - z} d\mu \quad \left(\frac{1}{\mu - z} = \frac{1}{\mu - z_0 - (z - z_0)} = \frac{1}{\mu - z_0} \cdot \frac{1}{1 - \left(\frac{z - z_0}{\mu - z_0}\right)} \right)$$

$$= \frac{1}{2\pi i} \cdot \int_C f(\mu) \cdot \sum_{n=0}^{\infty} \frac{1}{(\mu - z_0)^{n+1}} \cdot (z - z_0)^n d\mu$$

$$= \sum_{n=0}^{\infty} \left(\frac{1}{2\pi i} \cdot \int_C \frac{f(\mu)}{(\mu - z_0)^{n+1}} d\mu \right) \cdot (z - z_0)^n$$

$$= \sum_{n=0}^{\infty} \frac{f^{(n)}(z_0)}{n!} \cdot (z - z_0)^n$$

Corollary, Liouville's Theorem.

If $f: \mathbb{C} \rightarrow \mathbb{C}$ is entire and bounded, then it is constant.

Proof. Given $z_0 \in \mathbb{C}$ and $R > 0$, $|f'(z_0)| \leq \frac{1}{R} \cdot \sup_{C_R} |f| \rightarrow 0$ as $R \rightarrow \infty$. Therefore, $f' = 0$.

Thus f is constant.

Fundamental Theorem of Algebra.

Non-constant polynomial $P(z) = a_n z^n + a_{n-1} z^{n-1} + \dots + a_0 \in \mathbb{C}[z]$ has a root in $\mathbb{C}$.

Proof. Suppose that $P(z)$ has no root in $\mathbb{C}$. Then, $1/P(z)$ is well-defined on $\mathbb{C}$, entire.

observe that
$$\frac{P(z)}{z^n} = a_n + \left(\frac{a_{n-1}}{z} + \dots + \frac{a_0}{z^n} \right) \quad (z \neq 0).$$

The terms $a_{n-k} \cdot z^{-k}$ ($k=1, \dots, n$) converges to zero as $|z| \rightarrow \infty$, thus for some $R > 0$,

$$\begin{aligned} |P(z)| &\geq 2|a_n| \cdot z^n \quad (|z| > R) \\ &\geq 2|a_n| \cdot R^n \end{aligned}$$

Therefore it is bounded below in $|z| > R$, and clearly in $|z| \leq R$.

Now, $\frac{1}{P}$ is bounded, thus constant. Contradiction.